



Introduction to Data Science

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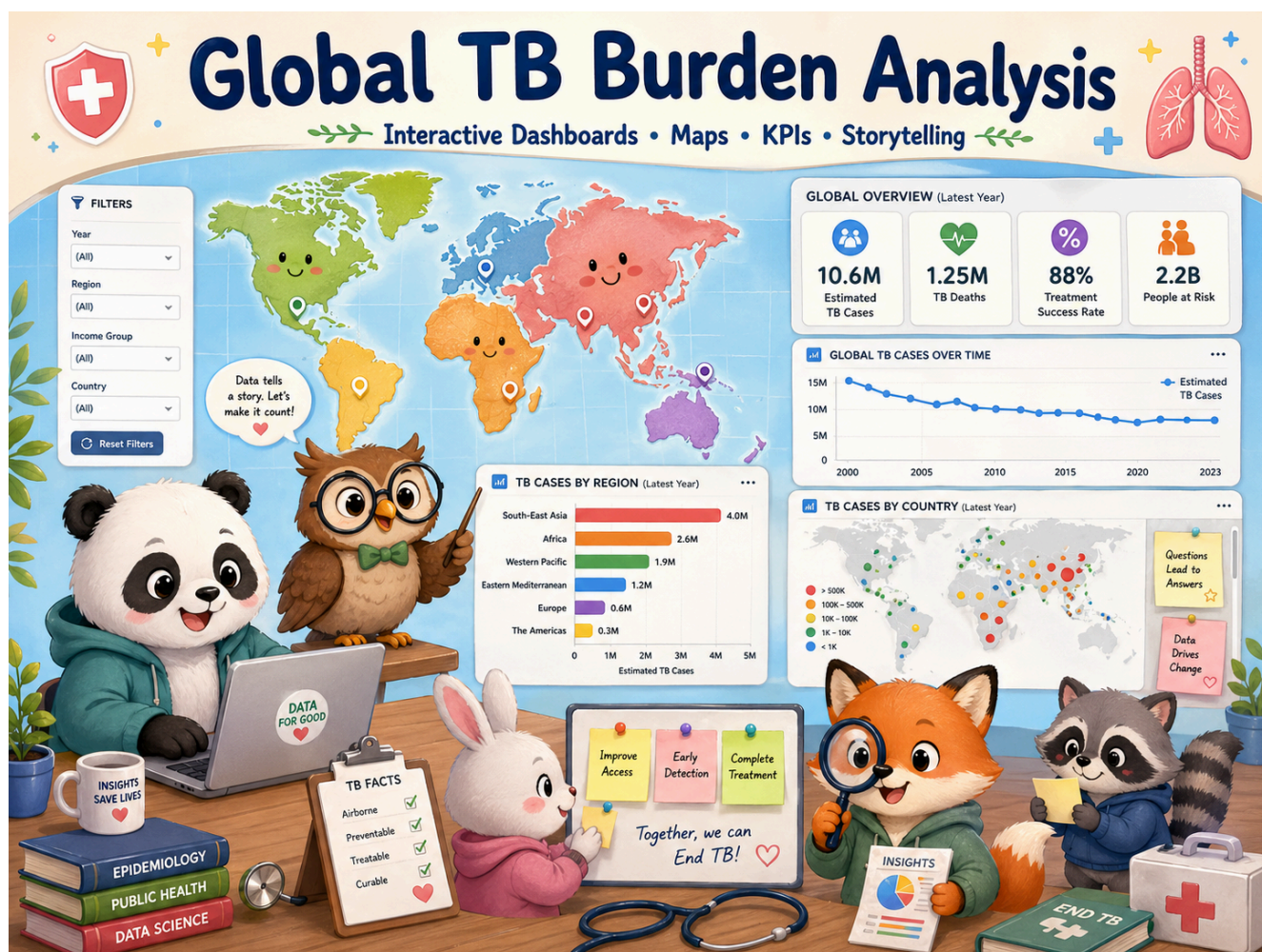
Assignment 1

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Deadline: 21st Ordibehesht

Introduction

In this assignment, students will apply key data science concepts through three practical tasks: sampling and inference, Tableau dashboard storytelling, and Power BI dashboard improvement. They will generate and compare samples from a 2D Gaussian distribution using direct sampling and Langevin Dynamics, analyze global Tuberculosis data through interactive Tableau dashboards, and enhance a financial Power BI dashboard using advanced indicators and better visual design. Overall, the assignment connects statistical reasoning, visualization, and business intelligence skills in practical data analysis settings.



Task 1: Sampling (35 Points)

In this assignment, you will study how samples are generated from a probability distribution and how those samples can be used for statistical inference. You will work with a known 2D Gaussian distribution. First, you will visualize the distribution. Then, you will generate samples using two different methods. (Direct sampling, and Langevin Dynamics sampling)

1. Define and Visualize the Distribution

Define the 2D Gaussian distribution with: $\mu = [0, 0]$, $\Sigma = I$

Then create a 2D grid of points and compute the probability density value at each grid point. Finally, visualize the density using a heatmap or contour plot.

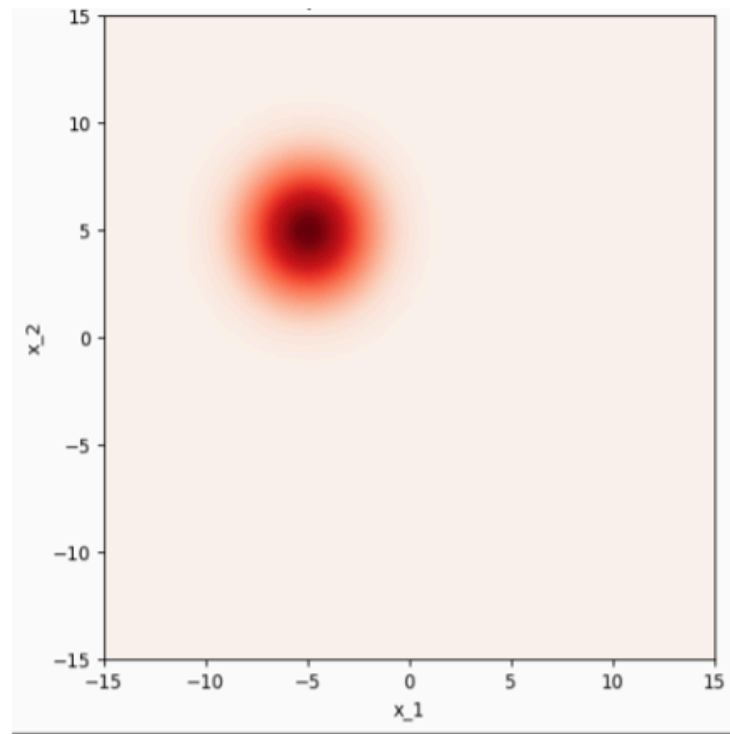


figure1:Heatmap of 2D gaussian

Include the following in your report:

1. A heatmap or contour plot of the target distribution.
2. A short explanation of the plot.

Questions:

1. Why is the distribution symmetric?
2. What role does the covariance matrix play in the shape of the distribution?

2. Sampling Methods

In this part, you will generate samples from the target distribution using two methods.

Generate 1000 samples from the target distribution using NumPy's built-in sampler as a reference method. (Use the mean and covariance defined in section 1.)

Include the following in your report:

1. A scatter plot of the 1000 direct samples.

Langevin Dynamics is a sampling method that generates samples from a target distribution by combining deterministic movement with randomness. The key idea is to iteratively move points toward regions of higher probability while also adding noise to ensure proper exploration of the space. At each step, the algorithm uses information about the shape of the distribution to guide movement, while randomness prevents the process from getting stuck and allows it to explore different regions. The main tool used in Langevin Dynamics is the score function, defined as the gradient of the log-density:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x})$$

Intuitively, the score function tells us in which direction the probability density increases most rapidly, meaning it points toward regions where the distribution assigns higher probability. The magnitude of this vector indicates how strongly the density changes in that direction. An important property of the score function is that it does not depend on the normalization constant of the distribution, which makes it especially useful when the exact probability density is not fully known. For the standard Gaussian distribution used in this part, the score function has the form:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = -\mathbf{x}$$

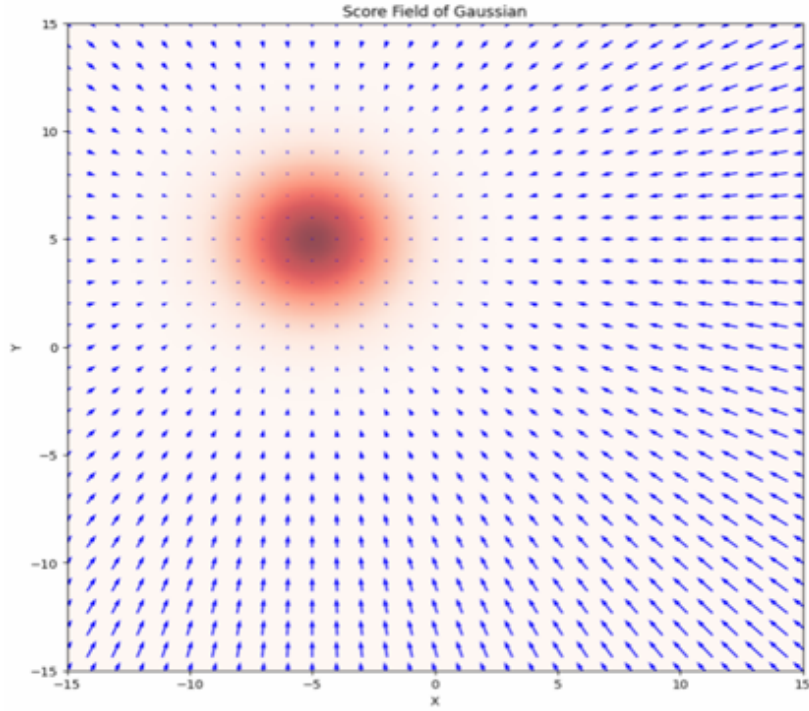


Figure 2: score field of distribution

The update rule of Langevin Dynamics is given by:

$$x_{t+1} = x_t + \epsilon \nabla_x \log p(x) + \sqrt{2\epsilon} z_t$$

where:

- x_t is the current point
- ϵ is the step size
- $z_t \sim N(0, I_2)$ is Gaussian noise

In this equation, the term involving the score function moves the current point toward more likely regions, while the noise term introduces randomness drawn from a standard normal distribution. The parameter controls the size of each step, affecting both how quickly the samples move and how much randomness is added. By repeating this update multiple times, the generated points gradually form a set of samples that approximates the target distribution.

1. Implement the score function.
2. Initialize 1000 independent particles from $x_0 \sim \text{Uniform}([-5, 5]^2)$ for $i = 1, \dots, 1000$
3. Run Langevin Dynamics independently for each particle for $T=500$ iterations. Use the final states $\{x_T^{(i)}\}_{i=1}^{1000}$ as the Langevin samples.

4. Store the final samples.
5. Track the trajectories of at least 10 sample points.
6. Plot the trajectories.
7. Plot the final sampled points.

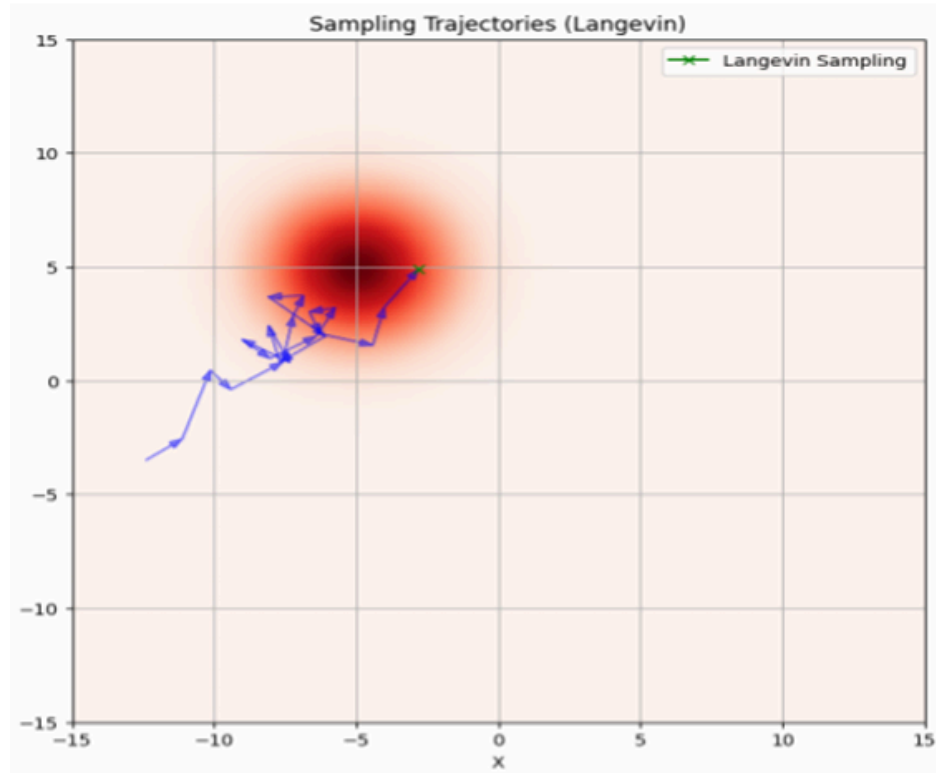


Figure 3:sampling trajectories

Include the following in your report:

1. How does changing the step size affect convergence speed and stability of the sampler?
2. How many iterations are required for the sampler to reach a stable distribution for your chosen step size? Explain how you determined this.
3. Compare the scatter plot with the density contours and explain why more samples appear near the origin.
4. If you remove the noise term from the update rule, what behavior do you expect and why?
5. Do all trajectories converge to similar regions of the space? What does this indicate about the sampler?

3. Inference from Samples

Use the generated samples to estimate statistical quantities and compare them with the true values. You must perform this task for both: Direct samples, and Langevin samples

For each method, estimate:

- Mean of dimension 1
- Mean of dimension 2
- Variance of dimension 1
- Variance of dimension 2

To estimate these quantities, use the generated samples from each method. Suppose you have N samples $\{x^{(1)}, x^{(2)}, \dots, x^{(N)}\}$, where each sample is a 2D vector. The mean of each dimension is computed by averaging the corresponding coordinate across all samples. The variance of each dimension is computed based on the deviations of that coordinate from its sample mean. Make sure that you compute these estimates separately for each dimension and for each sampling method, and use the same number of samples in both cases to ensure a fair comparison.

1. Why can the Langevin estimates still be biased even when the final samples come from independent chains?
2. How do the number of iterations T and the step size ϵ affect the accuracy of the Langevin samples?

Focus on whether each method produces samples that are centered near the true mean and have variance close to the true variance.

4. Compare the Sampling Methods

Compare direct sampling and Langevin Dynamics using the results from a single run.

Include the following in your report:

1. Estimated mean of each dimension for both methods.
2. Estimated variance of each dimension for both methods.
3. Absolute error of the estimated means compared with the true mean 0.
4. Absolute error of the estimated variances compared with the true variance 1.
5. A short discussion of which method better matches the target distribution.
6. If Langevin performs worse than direct sampling, explain whether this may be due to insufficient iterations, poor step size, or lack of convergence.

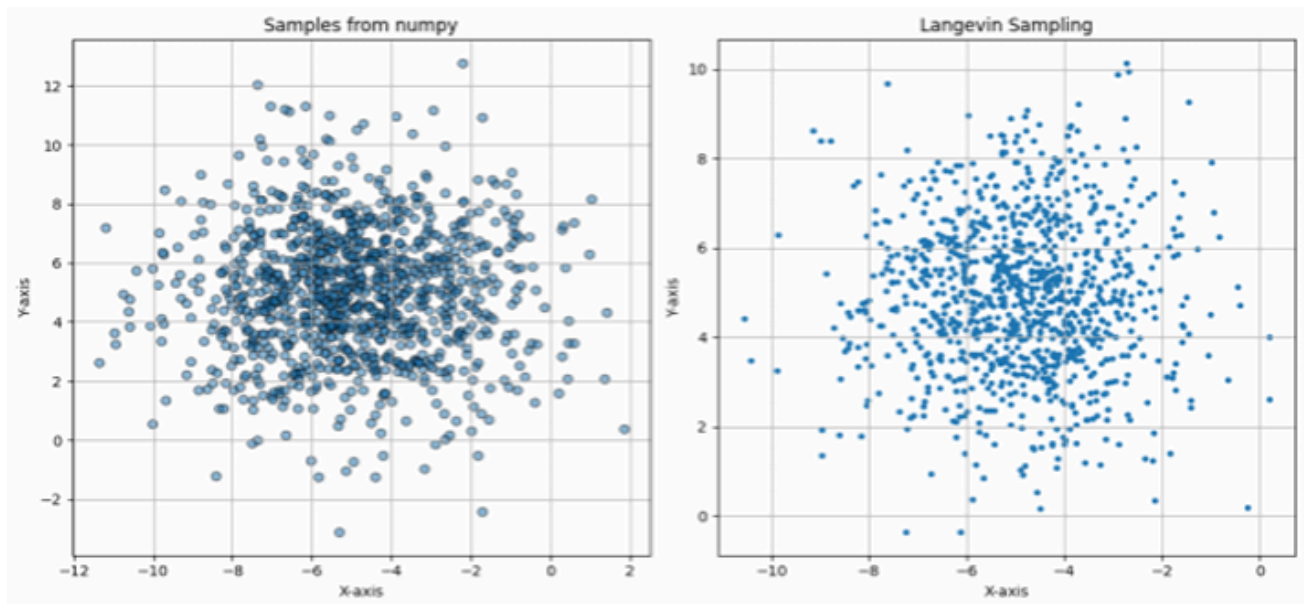


Figure 4: comparing samples

Analyze whether the Langevin sampler converges. Use:

1. Trajectory plots of selected particles.
2. Trace plots of the coordinates over time.
3. A plot of the average distance from the origin over iterations.

Questions:

1. Do the particles move toward the high-density region near the origin?
2. After approximately how many iterations do the trajectories appear stable?

Compare the sampled distributions with the true distribution. Use the following visualizations:

1. Scatter plot over density contours
2. 2D histogram and side-by-side comparison of direct and Langevin samples

Task 2: Tableau (35 Points)

In this task, you will utilize Tableau to analyze global Tuberculosis (TB) burden data and create a data-driven story through multiple interactive dashboards. The objective is to build a well-structured narrative using several dashboards, each focusing on different aspects of the data such as global trends, regional comparisons, and country-level insights.

Before starting Task 2, you may download Tableau using the provided [link\(windows\)](#) and [link\(Mac\)](#) for easier access. If preferred, you can also download it directly from its [official website](#), depending on your internet availability.

1. Dataset Details

1) TB_Burden_Country.csv

This file contains country-level tuberculosis burden indicators by year.

The accompanying .xlsx file column_descriptions.xlsx provides the column names and their corresponding descriptions and is included solely to clarify the meaning of the columns in this file.

2) Region.csv

This file contains country-to-region classification. It includes:

- Country or territory name: Official name of the country or territory.
- ISO 2-character code: Two-letter ISO country code (e.g., US, IN).
- ISO 3-character code: Three-letter ISO country code (e.g., USA, IND).
- ISO numeric code: Three-digit ISO numeric country code.
- Region: Countries are grouped into WHO-style regions.

This dataset is used to add regional context to each country.

3) Coordinate.csv

This file contains geographic location data for countries. It includes:

- Country or territory name: Official name of the country or territory.
- Latitude: The geographical latitude of the country.
- Longitude: The geographical longitude of the country.

This dataset enables map visualizations and geographic analysis in Tableau.

2. Data Preparation

- Import all datasets into Tableau:
TB_Burden_Country.csv, Region.csv, and Coordinate.csv.
- Join the datasets properly using country identifiers such as:
 - Country or territory name
 - ISO 2-character country code
 - ISO 3-character country code
 - ISO numeric country code
- After joining, each country record must contain:
 - TB indicators
 - Region information
 - Geographic coordinates (Latitude and Longitude)
- Handle null values if needed to ensure correct mapping and analysis.

3. Multiple Dashboards & Storytelling

Instead of creating only one dashboard, create multiple dashboards that together tell a clear and structured story about global TB trends.

- Create at least three dashboards , each focusing on different aspects of the analysis.
- Include interactive elements such as filters or parameters to enable dynamic exploration.
- Use KPIs effectively to highlight key performance metrics in the analysis

4. Insights & Storytelling Structure

Your story should be structured as:

- Introduction: Define the key focus of the analysis.
- Findings across multiple dashboards:
 - Each dashboard should highlight different aspects of the story.
 - Example insights: global TB incidence trends over time, regional differences in TB burden, countries with the highest estimated incidence or mortality, and geographic patterns in TB burden.
- Conclusion: Summarize the main findings and patterns seen in all the dashboards.

If meaningful insights are found, highlight them. If not, summarize the main trends in a

structured way. You do not need to prove the only one relationship, in addition to the lack of an event that is logical.

5. Design Best Practices

- Follow Gestalt Principles to keep the dashboards structured and easy to read.
- Use preattentive attributes like color, size, and position to highlight key insights.

6. Presentation & Submission

Submit a zip file that includes:

- A PDF report that includes:
 - A short written summary explaining the story behind the dashboards.
 - Screenshots of all dashboards/story pages.
 - A structured narrative highlighting key insights and trends.
 - A list of KPIs, filters, slicers, and parameters used in the report.

On presentation day, all group members must be prepared to explain:

- the structure of their dashboards,
- the data preparation steps,
- and how each report page contributes to the overall story.

7. Evaluation Criteria

- **Data Integration** – Properly joining and preparing datasets.
- **Dashboard Quality** – Well-structured, interactive, and connected dashboards.
- **Visualization Variety** – At least three dashboards with multiple types of visualizations.
- **Storytelling** – A single, structured narrative that connects all dashboards.
- **Use of Interactivity** – Meaningful filters, parameters, KPIs and dashboard actions.
- **Design Principles** – Applying Gestalt principles and preattentive attributes.

Task 3: Dashboard Improvement (35 Points)

Before starting the project and diving into its details, you need to install Power BI. Since it is developed by Microsoft, there is only a Windows version available. You can install it using this [link](#).

1. Project Overview

This project challenges you to become a financial data analyst and BI developer. You will start with a pre-built Power BI dashboard connected to a CSV file containing financial time series data for a stock. Your mission is to first thoroughly understand the existing dashboard and its underlying data. Then, leveraging your data science knowledge, you will enhance the dashboard by implementing more sophisticated analytical features and improving its overall effectiveness. Finally, you will present your findings and improvements.

2. Project Files

khodro_final_daily.csv: The raw financial time series data.

Pre-built Power BI Dashboard: A .pbix file that connects to the CSV and visualizes some of the data.

3. Become an expert

Your first task is to become an expert on the current dashboard:

1. **Data Source Exploration:** Locate the khodro_final_daily.csv file. Ensure Power BI can connect to it (you might need to adjust the data source path if it's hardcoded).
2. **Dashboard Deconstruction:**
3. **Purpose & Audience:** Who is this dashboard for? What questions does it aim to answer *now*?
4. **Data Model:** Examine the Power BI data model (Model View). Identify tables, relationships, and cardinality. Crucially, ensure a dedicated Date Table is present and correctly related to the date column from the khodro_final_daily.csv file. This is essential for all time-intelligence calculations.
5. **DAX Measures:** For *every* measure used in the dashboard, check its DAX code. Understand how it's calculated.
6. **Current Insights:** What key takeaways can be derived from the dashboard *as is*?
7. **Initial Weakness Identification:**
 - Based on your understanding, what are the current limitations?
 - Are there missing analytical elements common in financial analysis?
 - Is the design intuitive? Are there performance issues?

4. Enhancing the Dashboard

Leverage your data science course knowledge to transform this dashboard into a professional analytical tool. You are required to:

1. Implement Advanced Analytics: Apply at least three technical indicators or statistical methods using DAX (e.g., Moving Averages, Volatility analysis, RSI thresholding, or predictive trends). Note that there is no need for prediction we just want visualization of the calculated metrics in this project.
2. Elevate UX/UI: Redesign the dashboard layout to be intuitive, add dynamic tooltips, and ensure colors/visuals clearly communicate financial insights rather than just raw data.

Deliverables:

Send your original dashboard (.pbix file) along with a documentation of your work and findings in a pdf file :

- The Audit: Briefly summarize the original dashboard's weaknesses.
- The Solution: Detail the technical changes made. Include the DAX code, explain your chosen methodology, and justify your design decisions.
- Insight Discovery: Demonstrate what new analytical questions can now be answered that the original dashboard could not.

Questions (5 Points)

1. Visualization Design Principles

- What are Gestalt Principles of Visual Perception?
- Why are Gestalt principles important in visualization design?
- What are the main Gestalt principles?

2. Dashboards and Storytelling

- What is a dashboard?
- What is the purpose of a dashboard?
- What makes a good dashboard?

3. Score Function and Sampling Bias in Langevin Dynamics

Suppose the target density is written as:

$$p(\mathbf{x}) = \frac{1}{Z} \tilde{p}(\mathbf{x})$$

where Z is an unknown normalization constant. Show mathematically why the score function:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x})$$

does not depend on Z . Explain why this property is important for Langevin Dynamics.

Even if the target distribution is correct, estimates from Langevin samples may be biased. Explain at least three reasons why this bias may occur. Relate your answer to your experiment.

4. Markov Dependence in Langevin Sampling

Direct samples from NumPy are independent, but Langevin samples are generated through a Markov chain. Explain why Langevin samples are generally dependent. How does this dependence affect: variance estimation, standard error, and confidence intervals.

5. The Role of Data Labels

In data visualization, decisions about whether or not to include labels for specific data points can be nuanced.

Explain the context in which *omitting* data labels is a valid and effective design decision. Conversely, describe the context where *directly labeling* data points (e.g., showing the exact value above a bar or on a line) is crucial for clarity. When does a label become “clutter,” and when does it transform into essential “clarity”?

Provide a specific example of a data visualization scenario where you would choose *not* to label individual data points, and another scenario where direct labeling is absolutely non-negotiable for effective communication.

6. Cognitive Load and Visual Simplicity

Different elements within a visualization have varying impacts on how easily a viewer can understand the information.

Identify a single design choice or “decluttering” step that you believe has the most significant impact on reducing cognitive load. Explain your choice by referencing principles like the Gestalt Principles or the concept of the “Data-Ink Ratio.”

How does minimizing visual “noise” allow the core message of the data to be perceived more quickly and accurately?

Notes

- Upload your work as a zip file in this format on the website: DS_CA1_[Std number].zip.

If the project is done in a group, include all of the group members’ student numbers in the name.

- If the project is done in a group, only one member must upload the work.
- We will run your code during the project delivery, so make sure your results are reproducible.
- We will also use your public Tableau dashboard, so ensure it remains available and is not deleted.